

Time Series and Forecasting

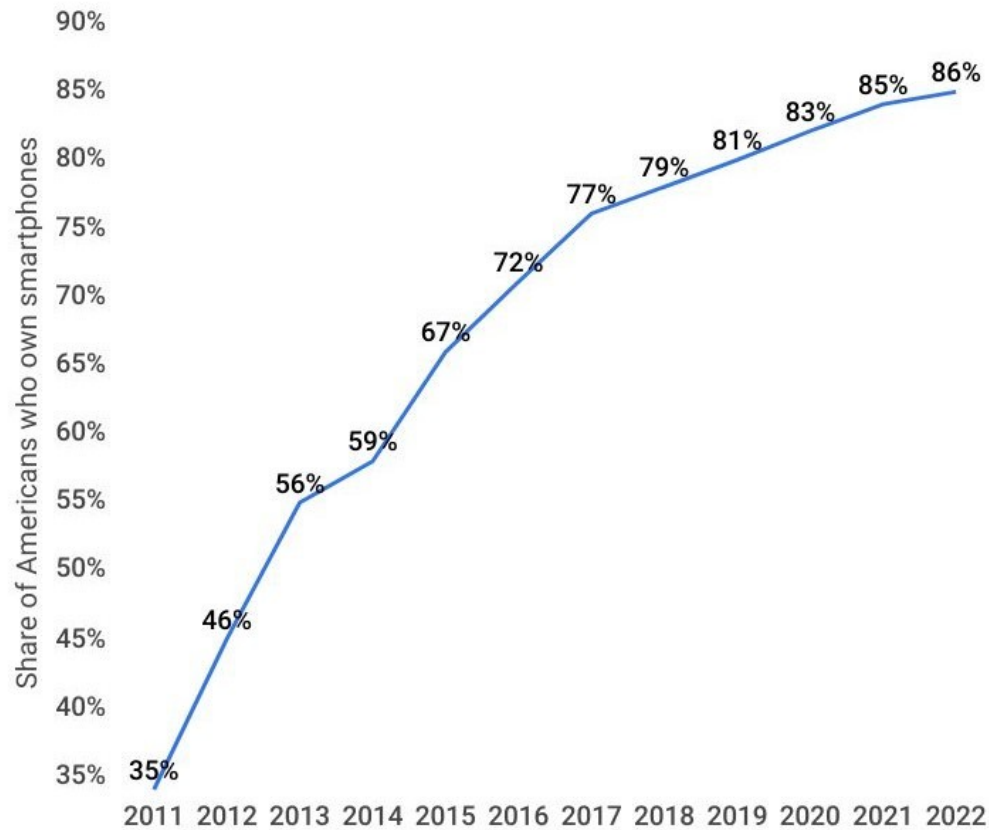
Management Science Principles

Time Series

Time Series Data

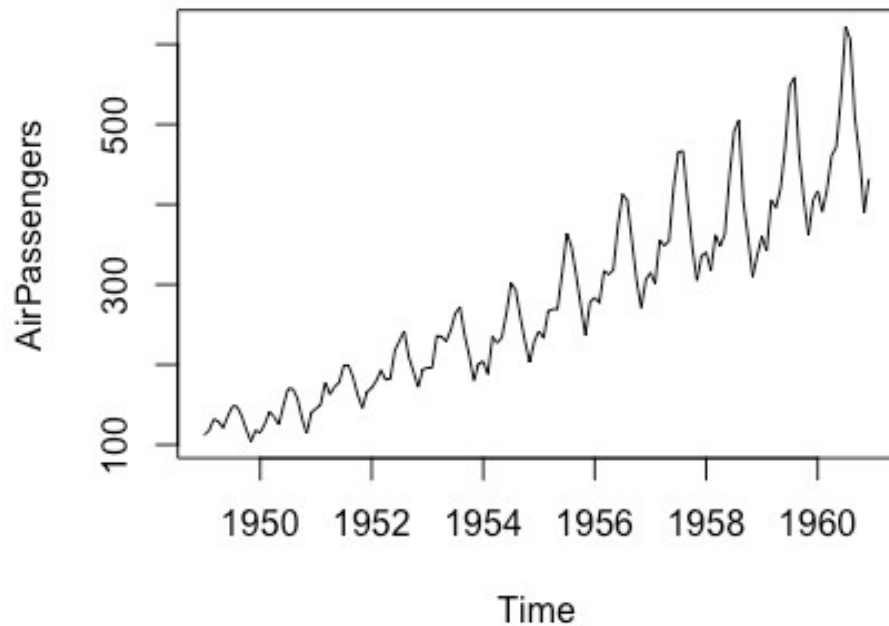
- Time series data consists of a collection of data points over a consistent interval.
- The term really applies to any type of information that is time-based.
 - Stocks and currencies
 - Real estate markets
 - Quarterly revenue
 - Temperature and weather
 - Populations
 - Sensor data
 - Just about anything measured by time...

US SMARTPHONE OWNERSHIP OVER TIME



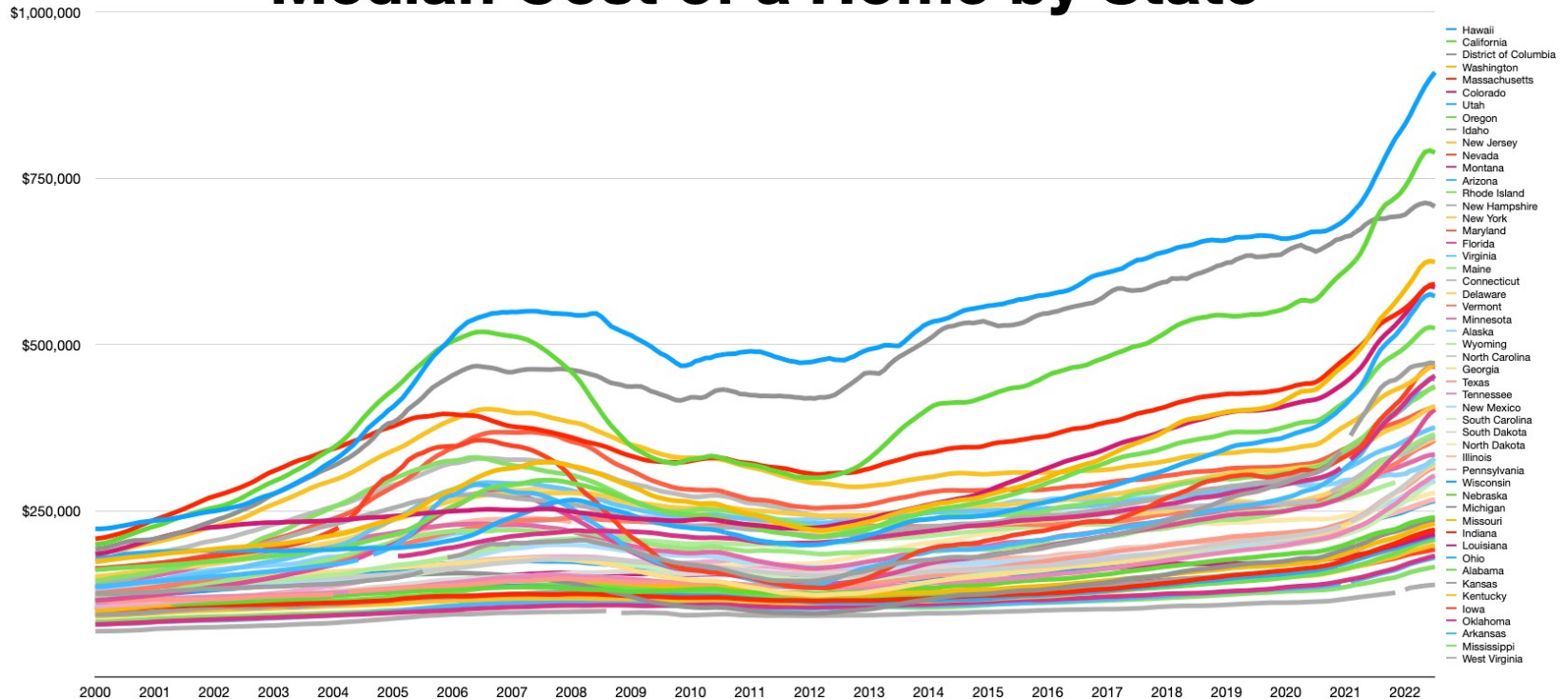
Credit: <https://www.zippia.com/advice/us-smartphone-industry-statistics/>

Air Passengers Dataset



Credit: <https://boostedml.com/2020/05/visualizing-time-series-in-r.html>

Median Cost of a Home by State



Credit: https://en.wikipedia.org/wiki/Timeline_of_the_2000s_United_States_housing_bubble



5 Key events



Nvidia Hits Record High as
Stock Surge Nears 120%
This Year



12.04% that day

Bloomberg · Nov 4, 2021

Nvidia Stock Jumps in After-
Hours Trading



14.02% that day

The Wall Street Journal · Feb 23...

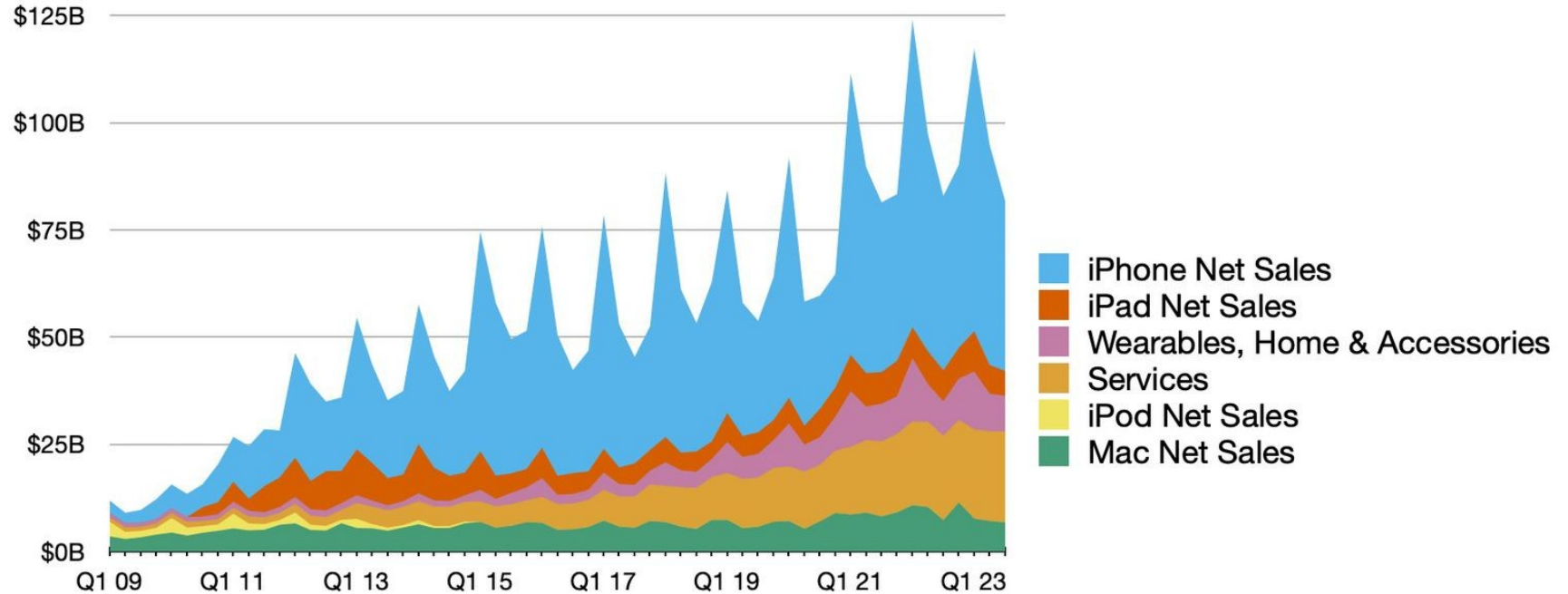
Nvidia shares surge after
strong fourth quarter results



16.4% that day

Yahoo News · Feb 22, 2024

Apple Inc. Revenue by Quarter



Credit: <https://www.macrumors.com/2023/08/03/apple-3q-2023-results/>

Activity Details

Wednesday 3 Dec

12:00a

11:59p

BIOMETRICS

✓ **HEART RATE**

65 avg ♡/min

✓ **STEPS**

6752 ⇅

5 avg ⇅/min

○ **CALORIES**

○ **SKIN TEMP**

○ **PERSPIRATION**

ACTIVITY

WALKING ✓

2886 ⇅

61 avg ⇅/min

41 min total

RUNNING ✓

0 ⇅

0 avg ⇅/min

0 min total

BIKING ✓

50 ⚡cal

6 avg ⚡/min

8 min total



Credit: <https://www.dcrainmaker.com/2016/02/understanding-continual-optical.html>

Uses of Time Series Data

- We work with time series all the time!
- Descriptive and explanatory analysis help us to interpret the meaning of and patterns in our data.
- Classification techniques rely on time series in applications like gesture recognition or speech processing.
- Trends help us to predict, or **forecast**, future values.

Attributes of Time Series

- Important attributes:

- Trend lines

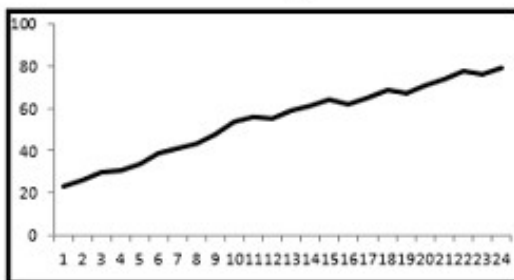
- Overall "smoothed" behavior of the data
- Can be linear or nonlinear

- Seasonality and Cyclicity

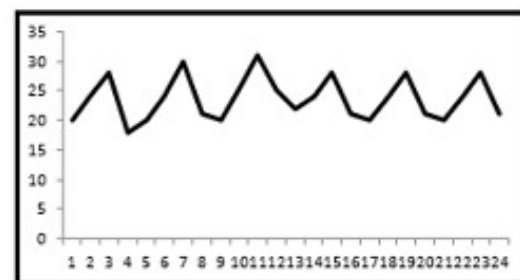
- Recurring patterns
- Seasonality is fixed intervals, and cyclicity is generic recurring behavior.

- Irregularity

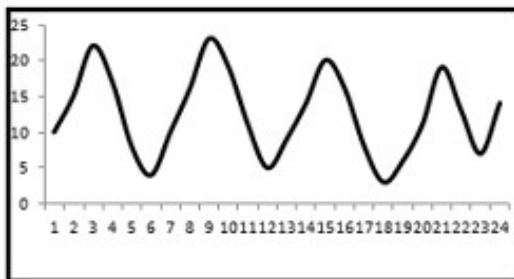
Components of Time Series



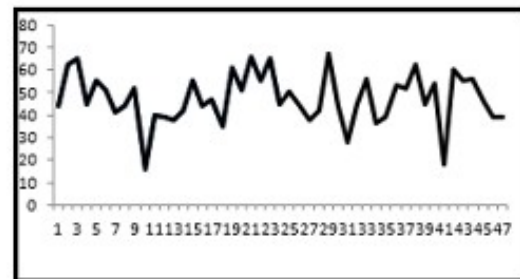
(a) Trend



(b) Seasonality



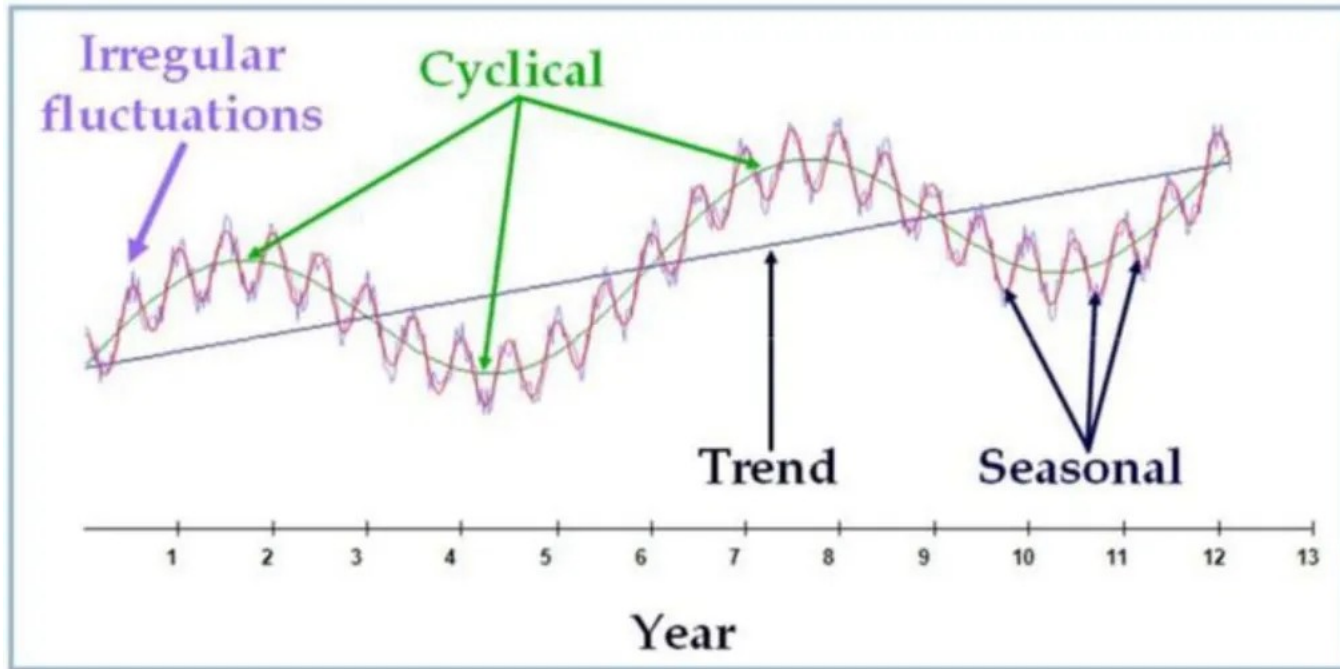
(c) Cyclicity



(d) Irregular

Attributes of Time Series

- A single dataset can exhibit multiple attributes!



Forecasting

Includes content adapted from
<https://www.youtube.com/@BrandonFoltz>

Forecasting

- Forecasting is the prediction of future values.
- In today's data-driven world, this means constructing models from historical data to predict the future.
- Common notation:
 - **t** is the time interval measured.
 - This could be years, quarters, months, weeks, or even minutes, seconds, etc.
 - **Y** is the actual data points.
 - Y_t is the value recorded at time t .
 - Y_{t+1} is the value recorded at time $t+1$.
 - **F** is the forecast value.
 - F_{t+1} is the predicted value at time $t+1$.
 - **e** is forecast error.
 - We measure error to evaluate and compare models.
 - e_t is the error of the forecast at time t (F_t) versus the actual value at time t (Y_t).

How to forecast?

We do this all the time!

Forecasting Example

- Consider this example: your sorority or fraternity hosts an event and 100 people attend.
- The second year you plan to host the event again and you design and advertise everything as before.
- What is a good rough estimate for how many people you might plan for this year?
- 100 people is a good rough estimate!
 - Note this ignores external factors you may have or already know like a potential increase in chapter size or event advertising campaigns.
 - This is just an example of the simplest model we could apply.

Naïve Forecasting

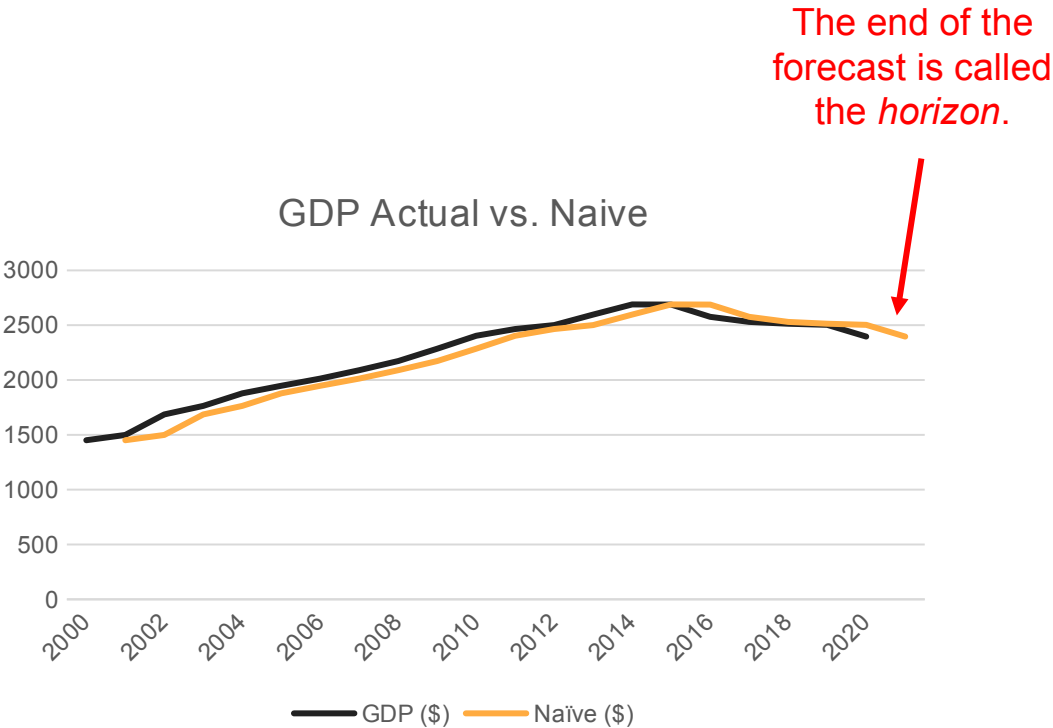
- The naïve forecasting model uses the current period value to predict the next one:
 - or you could say previous period to predict the current one.

$$F_{t+1} = Y_t$$

Year	GDP	Naïve
2000	1451	
2001	1499	1451
2002	1686	1499
2003	1764	1686
2004	1879	1764
2005	1948	1879
2006	2013	1948
2007	2090	2013
2008	2173	2090

Naïve Forecast Shifts the Time Series

Year	GDP	Naïve
2000	1451	
2001	1499	1451
2002	1686	1499
2003	1764	1686
2004	1879	1764
2005	1948	1879
2006	2013	1948
2007	2090	2013
2008	2173	2090
2009	2286	2173
2010	2404	2286
2011	2465	2404
2012	2501	2465
2013	2597	2501
2014	2689	2597
2015	2688	2689
2016	2576	2688
2017	2530	2576
2018	2513	2530
2019	2503	2513
2020	2396	2503
2021		2396



Forecast Error

- We want to calculate the error to evaluate model performance.
- This is like other error metrics we have seen.
 - E.g., mean absolute error (MAE), mean square error, and root mean square error (RMSE) – each could be a reasonable metric.

$$MSE = \frac{\sum_{t=1}^n (Y_t - F_t)^2}{n} = \frac{\sum_{t=1}^n e_t^2}{n}$$

Forecast Error Example

Year	GDP (\$)	Naïve (\$)	Error (\$)
2000	1451		
2001	1499	1451	=B3-C3



Year	GDP (\$)	Naïve (\$)	Error (\$)
2000	1451		
2001	1499	1451	48
2002	1686	1499	187
2003	1764	1686	78
2004	1879	1764	115
2005	1948	1879	69
2006	2013	1948	65
2007	2090	2013	77
2008	2173	2090	83
2009	2286	2173	113
2010	2404	2286	118
2011	2465	2404	61
2012	2501	2465	36
2013	2597	2501	96
2014	2689	2597	92
2015	2688	2689	-1
2016	2576	2688	-112
2017	2530	2576	-46
2018	2513	2530	-17
2019	2503	2513	-10
2020	2396	2503	-107
2021		2396	

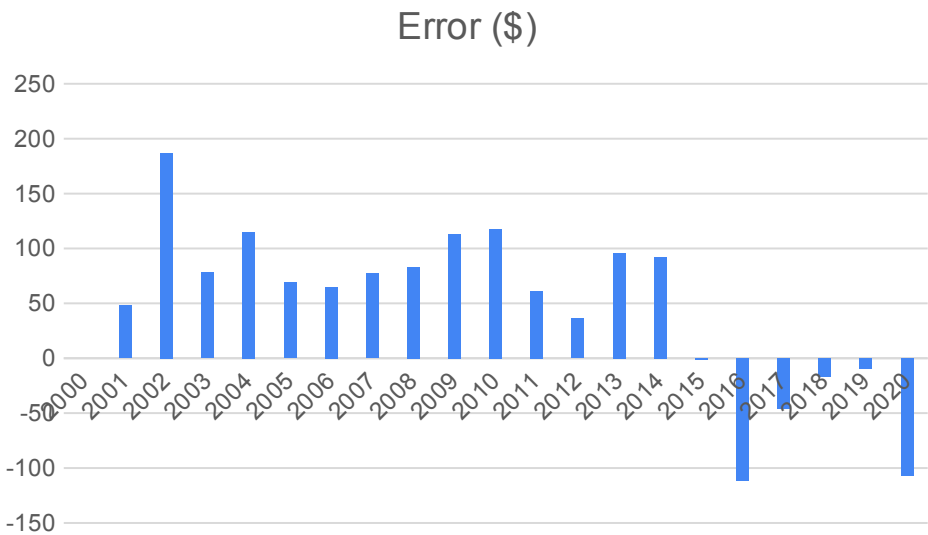
Forecast Error Example

Year	GDP (\$)	Naïve (\$)	Error (\$)	MSE
2000	1451			7713.75
2001	1499	1451	48	RMSE \$ 87.83
2002	1686	1499	187	
2003	1764	1686	78	
2004	1879	1764	115	
2005	1948	1879	69	
2006	2013	1948	65	
2007	2090	2013	77	
2008	2173	2090	83	
2009	2286	2173	113	
2010	2404	2286	118	
2011	2465	2404	61	
2012	2501	2465	36	
2013	2597	2501	96	
2014	2689	2597	92	
2015	2688	2689	-1	
2016	2576	2688	-112	
2017	2530	2576	-46	
2018	2513	2530	-17	
2019	2503	2513	-10	
2020	2396	2503	-107	
2021		2396		

←

MSE

=mean(power(D3:D22,2))



Note: Other error measures include different ways to scale, even by Naive

- Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{\sum_{t=1}^n \left| \frac{Y_t - F_t}{Y_t} * 100 \right|}{n}$$

- Symmetric Mean Absolute Percentage Error (sMAPE)

$$sMAPE = \frac{1}{n} \sum_{t=1}^n \frac{|Y_t - F_t|}{\frac{(Y_t + F_t)}{2}}$$

- Mean Absolute Scaled Error (MASE)

$$MASE = \frac{\sum_{t=1}^n |Y_t - F_t|}{n} * \frac{1}{\text{Naive Forecast MAE}}$$

Thus, the MASE for Naïve forecasting model is 1.

More Forecasting

- Let's revisit the example of an event you are planning.
- Now, imagine this is an annual event, and you want to use the last 3 years' attendance to estimate the current year's expected attendance.
- The last 3 years were 84, 112, and 95 attendees.

- What to do?
- Let's take the average!
 - $(84+112+95)/3 = 97$

Moving Average

- We call this model a **moving average**.
- The forecast is made by averaging over a period or “window” of time.
- Note: This averaging operation will have a “smoothing” effect.

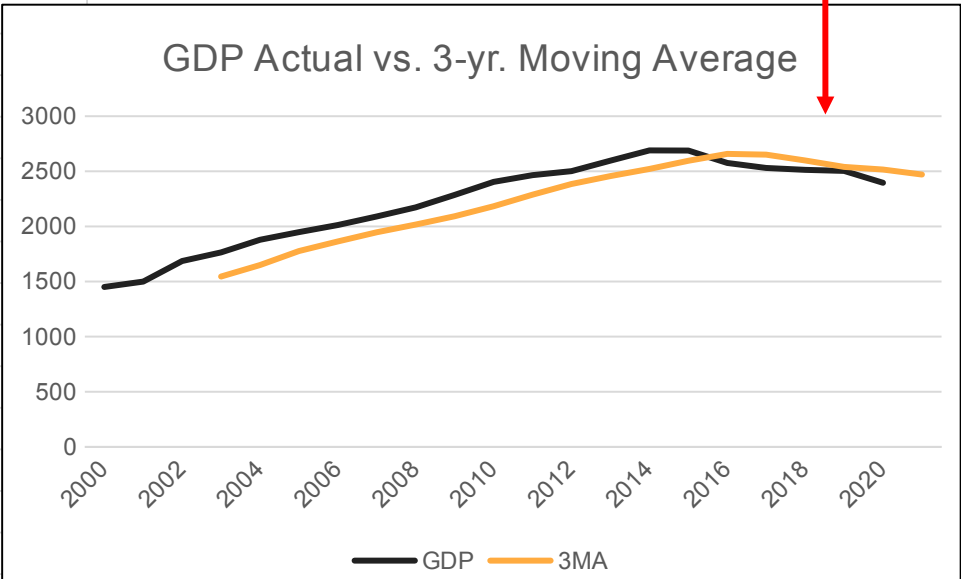
2000	1451
2001	1499
2002	1686
2003	1764 =MEAN(B2:B4)

Because we are forecasting, we are attempting to guess the next year. Thus, a 3-yr. window consists of the *previous 3 years*.

3-Year Moving Average Example

Year	GDP	Naïve Error	3MA	3MA Error	3MA MSE
2000	1451				23836.506
2001	1499	48			3MA RMSE
2002	1686	187			\$ 154.39
2003	1764	78	1545.3333	218.66667	
2004	1879	115	1649.6667	229.33333	
2005	1948	69	1776.3333	171.66667	
2006	2013	65	1863.6667	149.33333	
2007	2090	77	1946.6667	143.33333	
2008	2173	83	2017	156	
2009	2286	113	2092	194	
2010	2404	118	2183	221	
2011	2465	61	2287.6667	177.33333	
2012	2501	36	2385	116	
2013	2597	96	2456.6667	140.33333	
2014	2689	92	2521	168	
2015	2688	-1	2595.6667	92.333333	
2016	2576	-112	2658	-82	
2017	2530	-46	2651	-121	
2018	2513	-17	2598	-85	
2019	2503	-10	2539.6667	-36.666667	
2020	2396	-107	2515.3333	-119.33333	
2021			2470.6667		

Notice the new forecast series is slightly smoothed.



Weighted Moving Average

- Moving average is a very common method, and there are many enhancements to it.
- One such enhancement is the weighted moving average.
- In this model, we assign a weight to each input denoting its importance to the forecasted result.
- In the simplest case, we just assign equal weight to each input.
 - Note: this is actually just the basic form of moving average!

2000	1451		
2001	1499		
2002	1686		
2003	1764	$= 1/3 * B2 + 1/3 * B3 + 1/3 * B4$	

Selecting Weights

- We can just assign weights depending on how we want to build our model.
- Ideally, we could select the *optimal* weights to get the best model.
- What could we do?
 - Optimization!

2000	1451		
2001	1499		
2002	1686		
2003	1764	=?*B2+?*B3+?*B4	

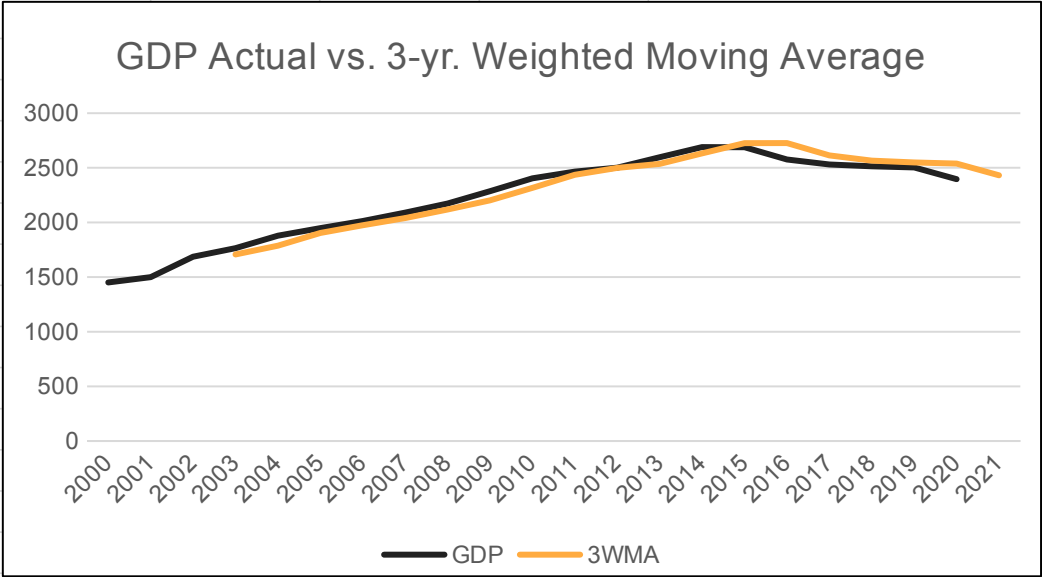
We will add one constraint for the standard weighted moving average model, which is that all weights must sum up to 1:

$$\sum_i w_i = 1$$

Optimized Weights using Gradient-based Nonlinear Optimizer

Year	GDP	Naïve Error	3WMA	Error	Weights		Sum of Weights
2000	1451				0.000	0.000	1.000
2001	1499	48					
2002	1686	187			3WMA MSE		
2003	1764	78	1686.00152	77.9984805	6499.97186		
2004	1879	115	1764.00159	114.99841	3WMA RMSE		
2005	1948	69	1879.00169	68.9983065	\$ 80.62		
2006	2013	65	1948.00176	64.9982444			
2007	2090	77	2013.00181	76.9981858			
2008	2173	83	2090.00188	82.9981164			
2009	2286	113	2173.00196	112.998042			
2010	2404	118	2286.00206	117.99794			
2011	2465	61	2404.00217	60.9978334			
2012	2501	36	2465.00222	35.9977784			
2013	2597	96	2501.00225	95.997746			
2014	2689	92	2597.00234	91.9976594			
2015	2688	-1	2689.00242	-1.0024235			
2016	2576	-112	2688.00242	-112.00242			
2017	2530	-46	2576.00232	-46.002322			
2018	2513	-17	2530.00228	-17.00228			
2019	2503	-10	2513.00226	-10.002265			
2020	2396	-107	2503.00226	-107.00226			
2021			2396.00216				

The result in this case suggests that a naïve prediction may be best!



Weighted Moving Average Error

- The weighted moving average model can forecast well when there is a trend in the data.
- While it can outperform Naïve forecasting, there is a delay inherent in moving average models, seen with the “smoothing” effect on the trendline.

